

CHE 555 – Homework 1

Contents

1 Full Solution	2
2 Plot	11

1 Full Solution

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↪ =====
REGRESSION ANALYSIS REPORT

CHE 555 – Homework 1
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↪ =====

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↪ =====
PROBLEM 1
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↪ =====

SETUP
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Curve:  $y = 2x^2 + 3x + 1$ 
The distance from a point  $(x, y(x))$  to the origin is
 $D(x) = \text{sqrt}[x^2 + y(x)^2]$ 
Minimising  $D^2$  :
 $D^2(x) = x^2 + (2x^2 + 3x + 1)^2$ 
 $dD^2(x^*)/dx = 2x + 2(2x^2 + 3x + 1)(4x + 3) = 0$ 
 $2x + 2(2x^2 + 3x + 1)(4x + 3) = 0 \rightarrow 8x^3 + 22x^2 + 19x + 3 = 0$ 
The roots of this cubic equations were solved numerically using
↪ the Newton-Raphson method.

=====
Optimal x      = -0.36380427
 $y(x^*)$       =  $2(-0.363804)^2 + 3(-0.363804) + 1$ 
               = 0.17329428
 $D^2(x^*)$      =  $(-0.363804)^2 + (0.173294)^2$ 
               = 0.16238446
 $D(x^*)$        =  $\sqrt{0.16238446} = 0.40296955$ 

ANSWER
=====
Nearest point : (-0.363804, 0.173294)
Distance      : 0.402970

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PROBLEM 2
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Variables : x = side length of square base (cm)
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$h = \text{height (cm)}$

Volume constraint : $x^2 \cdot h = 1000 \rightarrow h = 1000 / x^2$

Surface area (no top lid) :

$$\begin{aligned} S(x) &= x^2 + 4 \cdot x \cdot h \\ &= x^2 + 4 \cdot x \cdot (1000/x^2) \\ &= x^2 + 4000/x \end{aligned}$$

ANALYTICAL SOLUTION

$$dS/dx = 2x - 4000/x^2 = 0$$

$$2x^3 = 4000$$

$$x^3 = 2000$$

$$x^* = 2000^{(1/3)}$$

$$d^2S/dx^2 = 2 + 8000/x^3 > 0 \text{ for all } x > 0$$

NUMERICAL VALUES

$$x^* = 2000^{(1/3)} = 12.599210 \text{ cm}$$

$$h^* = 1000 / x^{*2} = 1000 / 12.599210^2 = 6.299605 \text{ cm}$$

$$S^* = (12.5992)^2 + 4000/12.5992 = 476.220316 \text{ cm}^2$$

$$d^2S/dx^2|_{x^*} = 6.0000 > 0$$

ANSWER

Optimal base side length : 12.5992 cm

Optimal height : 6.2996 cm

Minimum surface area : 476.2203 cm²

↪

PROBLEM 3

↪

Minimise: $f(x_1, x_2) = 4x_1 - x_1^2 - 12$

Subject to:

$$g_1 : 25 - x_1^2 - x_2^2 = 0$$

$$g_2 : 10x_1 - x_1^2 + 10x_2 - x_2^2 - 34 \geq 0$$

$$g_3 : (x_1-3)^2 + (x_2-1)^2 \geq 0$$

$$x_1, x_2 \geq 0$$

VARIABLE COUNT

Total Variables : 2 (x_1, x_2)

DOF ANALYSIS

Each independent equality constraint reduces DOF by 1.

Equality constraints : 1

DOF = total variables - equality constraints = $2 - 1 = 1$

Number of independent variables : 1

One valid choice : $\{ x_1 \}$

EXPLANATION

E1 defines x_2 as a function of x_1 : $x_2 = \sqrt{(25 - x_1^2)}$ (with $x_2 \geq 0$).
 $\hookrightarrow 0$).

Once x_1 is chosen (satisfying $0 \leq x_1 \leq 5$), x_2 is fully determined.

The inequality constraints further restrict the feasible range of
 $\hookrightarrow x_1$

but do not reduce the number of independent variables.

$g_3 = (x_1 - 3)^2 + (x_2 - 1)^2 \geq 0$ is always satisfied (sum of squares).

ANSWER

Total variables : 2

Independent variables : 1

One possible choice : $\{ x_1 \}$

$\hookrightarrow$ **PROBLEM 4**

$\hookrightarrow$ **GIVEN**

Cost per barrel : $C(P) = 50 + 0.1P + 9000/P$ (\$/barrel)

Selling price : \$300 / barrel

P = production rate (barrels/day)

PART A – Minimise cost per barrel

$$dC/dP = 0.1 - 9000/P^2 = 0$$

$$P^2 = 9000 / 0.1 = 90000$$

$$P^* = \sqrt{90000} = 300 \text{ barrels/day}$$

$$d^2C/dP^2 = 18000/P^3 > 0 \text{ for } P > 0$$

Numerical check:

$$P^* = 300.0 \text{ barrels/day}$$

$$C(P^*) = 50 + 0.1(300.0) + 9000/300.0 = 110.00 \text{ \$/barrel}$$

$$d^2C/dP^2|P^* = 0.000667 > 0$$

PART B – Maximise daily profit

$$\text{Revenue per day : } R(P) = 300 \cdot P$$

$$\text{Total cost / day : } TC(P) = C(P) \cdot P = 50P + 0.1P^2 + 9000$$

$$\begin{aligned} \text{Profit / day : } \Pi(P) &= R - TC = 300P - 50P - 0.1P^2 - 9000 \\ &= 250P - 0.1P^2 - 9000 \end{aligned}$$

$$d\Pi/dP = 250 - 0.2P = 0$$

$$P^* = 250 / 0.2 = 1250 \text{ barrels/day}$$

$$d^2\Pi/dP^2 = -0.2 < 0$$

Numerical check:

$$P^* = 1250 \text{ barrels/day}$$

$$\text{Revenue} = 300 \times 1250 = \$375,000.00/\text{day}$$

$$\text{Total cost} = 50(1250) + 0.1(1250)^2 + 9000 = \$227,750.00/\text{day}$$

$$\text{Profit } \Pi(P^*) = \$375,000.00 - \$227,750.00 = \$147,250.00/\text{day}$$

ANSWERS

(A) Production minimising cost/barrel : $P = 300$ barrel/day $\rightarrow C =$
 $\hookrightarrow \$110.00/\text{barrel}$

(B) Production maximising daily profit: $P = 1250$ barrel/day $\rightarrow \Pi =$
 $\hookrightarrow \$147,250.00/\text{day}$

$\hookrightarrow$

PROBLEM 5

$\hookrightarrow$

DATA

x	y
10	1.00
20	1.26
30	1.86
40	3.31
50	7.08

MODELS TO FIT

Model 1 : $y = \exp(a + b \cdot x)$ [exp-linear]
Model 2 : $y = \exp(a + b \cdot x + c \cdot x^2)$ [exp-quadratic]
Model 3 : $y = a \cdot x^b$ [power law]

METHOD

Models 1 & 2 are linearised by taking $\ln y$ and fitting via OLS.
Model 3 is linearised as $\ln y = \ln a + b \cdot \ln x$ and fitted via OLS.
Full regression statistics (R^2 , adj- R^2 , SE, RMSE, F-test, t-tests, ANOVA table) are reported for each model in the sections below.

Regression details:

↪ _____
Model: $y = \exp(a + b \cdot x)$

↪ _____
Observations : 5
Parameters (p) : 2
DOF (error) : 3

— Goodness of Fit —
 R^2 : 0.955527
 R^2 (adjusted) : 0.940703
SE of regression : 0.192228
RMSE : 0.148900
SSE : 0.110855
SSR : 2.381815
SST : 2.492670

— Overall F-Test —
F-statistic : 64.4574
F p-value : 0.00403481

— Parameter t-Tests (H_0 : param = 0, $\alpha = 0.05$) —
Param Estimate Std Err t-stat p-value 95% CI
↪ lower 95% CI upper

↪ _____
a -0.662933 0.201611 -3.2882 0.046141
↪ -1.304549 -0.021317 *

b 0.048804 0.006079 8.0285 0.00403481
 ↳ 0.029458 0.068149 *
 (* p < 0.05)

— ANOVA Table —

Source	df	SS	MS	F
↳ p-value				
Regression	1	2.381815	2.381815	64.4574
↳ 0.00403481				
Residual	3	0.110855	0.036952	
Total	4	2.492670		

↳ —
 Model: $y = \exp(a + b \cdot x + c \cdot x^2)$

↳ —
 Observations : 5
 Parameters (p) : 3
 DOF (error) : 2

— Goodness of Fit —

R² : 0.999968
 R² (adjusted) : 0.999936
 SE of regression : 0.006314
 RMSE : 0.003993
 SSE : 0.000080
 SSR : 2.492591
 SST : 2.492670

— Overall F-Test —

F-statistic : 31265.9950
 F p-value : 3.19826e-05

— Parameter t-Tests (H₀: param = 0, α = 0.05) —

Param	Estimate	Std Err	t-stat	p-value	95% CI
↳ lower	95% CI upper				

↳					
a	-0.040266	0.013541	-2.9736	0.096929	
↳	-0.098528	0.017997			
b	-0.004568	0.001032	-4.4264	0.0474371	
↳	-0.009008	-0.000128	*		
c	0.000890	0.000017	52.7166	0.000359642	
↳	0.000817	0.000962	*		

(* p < 0.05)

— ANOVA Table —

Source	df	SS	MS	F
↪ p-value				
Regression	2	2.492591	1.246295	31265.9950
↪ 3.19826e-05				
Residual	2	0.000080	0.000040	
Total	4	2.492670		

↪
Model: $y = a \cdot x^b$ [log-log space]

↪
Observations : 5
Parameters (p) : 2
DOF (error) : 3

— Goodness of Fit —

R² : 0.818733
R² (adjusted) : 0.758311
SE of regression : 0.388089
RMSE : 0.300612
SSE : 0.451838
SSR : 2.040832
SST : 2.492670

— Overall F-Test —

F-statistic : 13.5502
F p-value : 0.0347317

— Parameter t-Tests (H₀: param = 0, α = 0.05) —

Param	Estimate	Std Err	t-stat	p-value	95% CI
↪ lower	95% CI upper				

ln a	-2.863028	1.010440	-2.8334	0.0660019	
↪	-6.078700	0.352644			
b	1.123962	0.305337	3.6811	0.0347317	
↪	0.152245	2.095680 *			

(* p < 0.05)

— ANOVA Table —

Source	df	SS	MS	F
↪ p-value				

Regression	1	2.040832	2.040832	13.5502
↪ 0.0347317				
Residual	3	0.451838	0.150613	
Total	4	2.492670		

Original-scale R^2 for Model: $y = a \cdot x^b$: 0.726703 (adj R^2 : 0.635604)

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MODEL COMPARISON SUMMARY

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Model		R^2	adj R^2	RMSE
↪ F-stat	F p-val			

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↪				
exp_linear		0.955527	0.940703	0.148900
↪ 64.4574	0.00403481			
exp_quadratic		0.999968	0.999936	0.003993
↪ 31265.9950	3.19826e-05			
powerLaw		0.726703	0.635604	0.300612
↪ 13.5502	0.0347317			

Best model (highest adj R^2): exp_quadratic

FITTED VALUES vs. DATA

x	y_obs	exp-lin	exp-quad	power
10	1.0000	0.8395	1.0030	0.7596
20	1.2600	1.3677	1.2513	1.6554
30	1.8600	2.2282	1.8650	2.6111
40	3.3100	3.6300	3.3210	3.6079
50	7.0800	5.9136	7.0651	4.6364

RESIDUALS

x	exp-lin	exp-quad	power
10	0.1605	-0.0030	0.2404
20	-0.1077	0.0087	-0.3954
30	-0.3682	-0.0050	-0.7511
40	-0.3200	-0.0110	-0.2979
50	1.1664	0.0149	2.4436

CONCLUSION

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The exp-quadratic model has the highest adjusted R^2
among the three candidates.
Best model: exp-quadratic

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END OF REPORT
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2 Plot

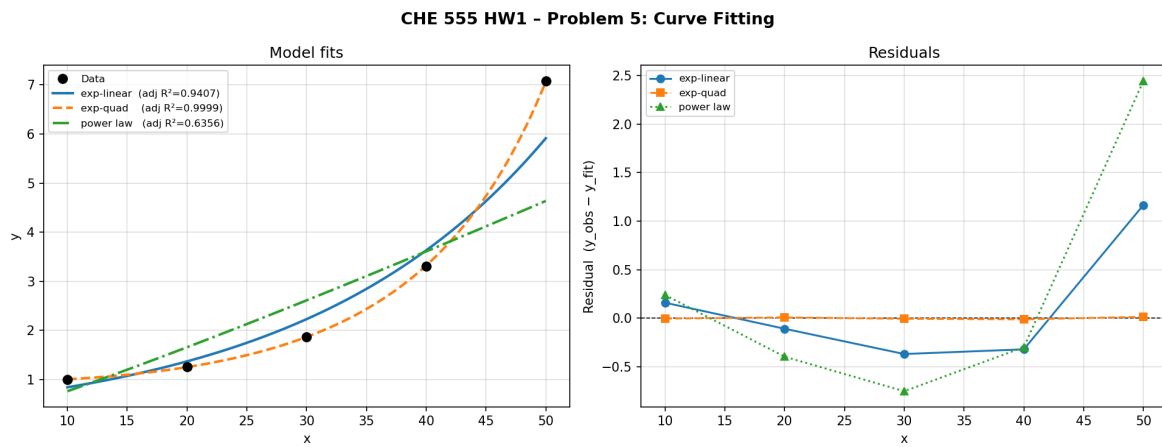


Figure 1: Curve fitting comparison for Problem 5.